Introduction to Scrum

Digital T-Level – Software Testing

Today

- Definition of Scrum & theory of Scrum
- The Scrum team
- Scrum events
- Scrum artefacts

The Scrum Guide™

- This presentation is based around the contents of the Scrum Guide.
- The guide is freely available at: http://www.scrumguides.org/
- The Scrum Guide was produced between the co-founders of the Scrum framework – Ken Schwaber & Jeff Sutherland

Definition of Scrum

- Scrum (n): A framework within which people can address complex adaptive problems, while productively and creatively delivering products of the highest possible value.

- Scrum is a process framework that has been used to manage work on complex products since the early 1990s.
- Scrum is not a process, technique, or definitive method.
- Rather, it is a framework within which you can employ various processes and techniques.

The Scrum Team

- The Scrum Team consists of:
  - A Product Owner
  - The development Team
  - A Scrum Master

- Scrum Teams are self organising, cross-functional teams.
- Self-organising teams choose how best to accomplish the work.
- Cross-functional teams have all of the competencies needed to accomplish the work.

The Scrum Team

- The Team model in Scrum is designed to optimise flexibility, creativity and productivity.
- Scrum Teams deliver products iteratively and incrementally, maximising opportunities for feedback.
- Incremental deliveries of a “Done” product ensure a potentially useful version of a working product is always available.

The Product Owner

- The Product Owner is the sole person responsible for managing the Product Backlog. This includes:
  - Clearly expressing Product Backlog items
  - Ordering the items in the product backlog to achieve best goals and missions.
  - Ensuring the Product Backlog is visible, transparent, clear to all and shows what the Scrum Team will work on next.
  - Ensuring the Development Team understands items in the product Backlog to the level needed.
- The product owner is one person – not a committee.

The Development Team

- The Development Team consists of professionals who do the work of delivering a potentially releasable increment of a “Done” project.
- They have the following characteristics:
  - They are self-organising.
  - They are cross-functional.
  - Scrum recognises no titles for Development Teams members other than developer.
  - Scrum recognises no sub-teams in the Development Team.
  - Individual team members may have specialised skills and areas of focus, but accountability belongs to the Development Team as a whole.

The Scrum Master

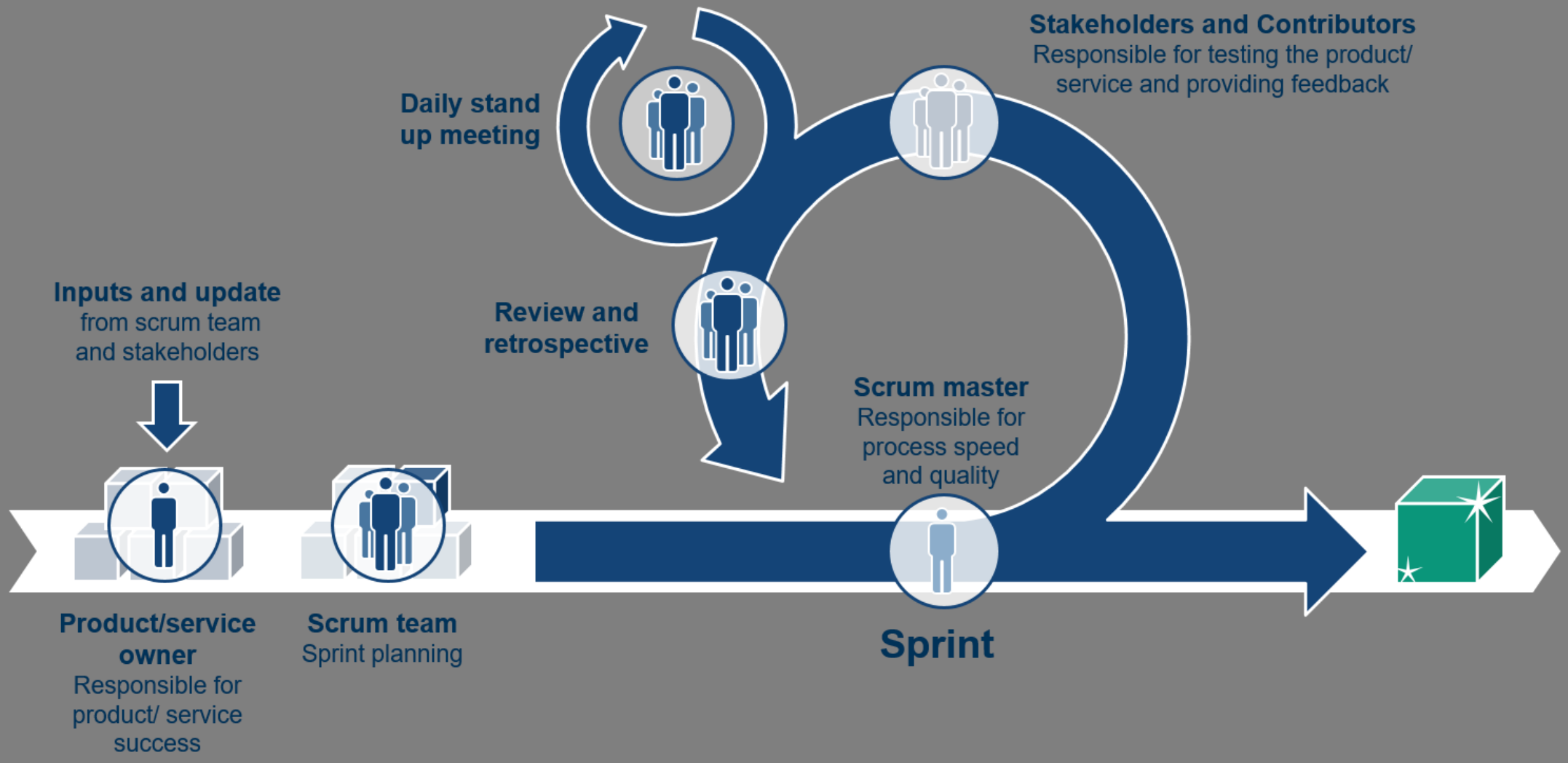
- The Scrum Master is responsible for ensuring Scrum is understood and enacted.
- Scrum Masters do this by ensuring that the Scrum Team adheres to Scrum theory, practices and rules.
- Scrum Master Serves…
  - Scrum Master service to the Product owner
  - Scrum Master service to the Development Team
  - Scrum Master service to the Organisation

Scrum Events

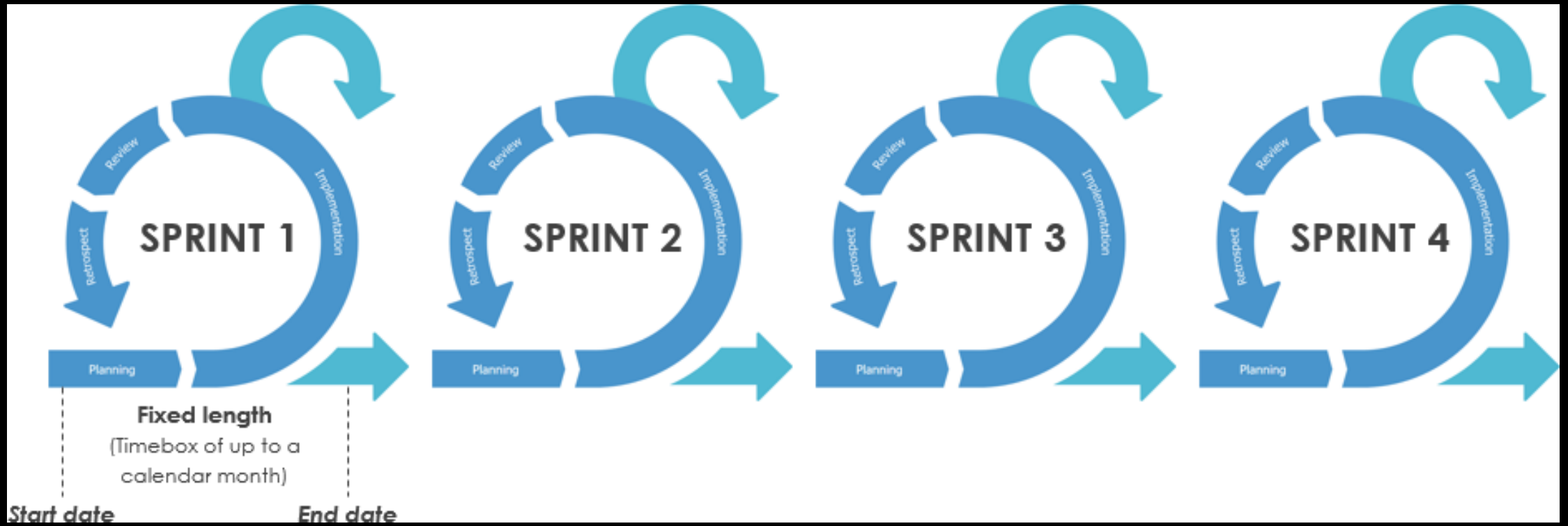
- Prescribed events are used in Scrum to create regularity and to minimise the need for meetings not defined in Scrum.
- All events are time-boxed events – every event has a maximum duration.
- Once a Sprint begins, its duration cannot be altered.

The Sprint

- The heart of Scrum is a Sprint
  - A time-box of one month or less…
  - …during which time a “Done”, useable and potentially releasable product increment is created.
  - Sprints best have consistent durations throughout a development effort.
  - A new Sprint starts immediately after the conclusion of the previous Sprint.



https://www.unity.de/fileadmin/Consulting_Products/Innovation-Scrum-Process-EN.png



https://www.visual-paradigm.com/servlet/editor-content/scrum/why-fixed-length-of-sprints-in-scrum/sites/7/2018/11/sprints-in-scrum.png

During the Sprint

- No changes are made that would endanger the Sprint Goal
- Quality goals do not decrease
- Scope may be clarified and re-negotiated between Product Owner and Development Team as more is learned.

- Each Sprint has a definition of what is to be built.
- Sprints are limited to one calendar month.

Cancelling a Sprint

- A sprint can be cancelled before the Sprint time-box is over.
- Only the Product Owner has the authority to cancel a Sprint.
- A Sprint would be cancelled if the Sprint Goal becomes obsolete.
- Sprint cancellations consume resources.
- Sprint cancellations are often traumatic to the Scrum Team, so…
- …they are very uncommon.

Sprint Planning

- The work to be performed in the Sprint is planned at the Sprint Planning.
- This plan is created by the collaborative work of the entire Scrum Team.
- Sprint planning is time-boxed to a maximum of eight hours for a one month sprint.
- Sprint Planning answers the following:
  - What can be delivered in the Increment resulting from the upcoming Sprint?
  - How will the work needed to deliver the Increment be achieved?

Sprint Goal

- The Sprint Goal is an objective set for the Sprint that can be met through the implementation of Product Backlog.
- It provides guidance to the Development Team on why it is building the Increment.
- It is created during the Sprint Planning meeting.

Sprint Review

- At the end of each sprint, there is a sprint review.

- During the review the product is demonstrated and…
- …reviewed in terms of satisfying the sprint goal.

- A sprint review is typically 1 to 2 hours in length and…
- …attended by the product owner, scrum master, development team and client.

Sprint Retrospective

- After the sprint review a retrospective is carried out.

- The retrospective looks back on the success of the sprint:
  - What problems occurred
  - How could future problems be avoided
  - What would help the next sprint.

- The review is focussed on the product and retrospective is concerned with the processes and how the sprint went.

Scrum Advantages and Disadvantages

- In the activity, you will need to compile a list of both positive and negative points of adopting a scrum approach to developing software projects.

- When completing this activity, remember to relate it to software testing.